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Summary: This report reviews the digital entrepreneurship policy that the governments of Korea, Japan, China, the United Kingdom, the United States, France and Germany set out in 2022 in six domains and analyses it in its context, against theory and in practice. Data, payments and platform power organised the year's binding measures: China's rules on data exports, Japan's stablecoin law, France's electronic invoicing law, to be phased in, and Germany's special abuse control of large platforms, while AI was governed by principles and voluntary frameworks. The best measures reached digital start-ups directly, as France's DeepNum20 programme did, or checked the platforms they depend on, as Germany's section 19a decisions did; the weakest were the United States' proclamation on culture and China's literacy aims for skills. Much of the year's policy was still a plan, bill or draft at its end; the United Kingdom will close its software voucher scheme and cut SME R&D relief, though it kept its rise in seed relief; and governments reported mostly outputs and claimed effects, with little evaluation.

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EEPR 2022

Digital Entrepreneurship Policy in Korea, Japan, China, the United Kingdom, the United States, France and Germany, 2022

IFEED Institute for Entrepreneurial Ecosystem Development

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Digital Entrepreneurship Policy in Korea, Japan, China, the United Kingdom, the United States, France and Germany, 2022

Abstract

This report reviews and analyses the digital entrepreneurship policy of the governments of the Republic of Korea, Japan, the People's Republic of China, the United Kingdom, the United States, France and Germany in their own plans, laws, budgets and announcements. The policies are reviewed in six domains — regulation and law, finance, culture, support, human capital and markets — and analysed in their context, against theories of digital ecosystems, platforms, data regulation, R&D tax relief, high-growth firms, entrepreneurial culture and immigrant talent, and in practice. Data, payments and platforms organised the year's binding measures. China's assessment of outbound data took effect on 1 September and its deep synthesis rules will apply from 10 January 2023; Japan enacted a stablecoin law; France enacted an electronic invoicing duty, to be phased in from a later date; and Germany's Federal Cartel Office placed Alphabet, Meta and Amazon under special abuse control, while platform rules awaited a bill in the United Kingdom and were under study in Japan. AI was governed by principles in Japan and the United Kingdom and by a non-binding blueprint in the United States. Support for SMEs' digital adoption ran in China, Korea, France, Germany and the United Kingdom, and France's DeepNum20 programme reached deep-tech digital start-ups, but the United Kingdom will close Help to Grow: Digital. The United States widened practical training for foreign graduates in cloud and data fields, the United Kingdom opened routes for graduates and scale-ups, and routes for founders were plans in Korea and Japan. The United Kingdom will cut its SME R&D relief but extend relief to data and cloud costs and keep larger seed relief, and governments reported mostly outputs and claimed effects, with little evaluation. The period covered is January to December 2022.

Keywords

data governance, outbound data transfers, AI governance, stablecoins, crypto-assets, platform competition, regulatory sandboxes, SME digitalisation, electronic invoicing, digital skills, skilled immigration

JEL Classification

L26, L86, O33, K24, L51, O38

Entrepreneurial Support Research Division, Institute for Entrepreneurial Ecosystem Development (IFEED).

1 INTRODUCTION

Key findings

- Digital entrepreneurship policy in 2022 was mostly strategies, plans and proposals; binding measures came on data, payments, invoicing and platform power.
- China's assessment of outbound data took effect on 1 September and its deep synthesis rules will apply from 10 January 2023; Japan enacted a stablecoin law, and AI was governed by principles and voluntary frameworks in Japan, the United Kingdom and the United States.
- Germany's Federal Cartel Office placed Alphabet, Meta and Amazon under special abuse control; platform rules awaited a bill in the United Kingdom and were under study in Japan.
- Support for SMEs' digital adoption ran in China, Korea, France, Germany and the United Kingdom, and France's DeepNum20 programme reached deep-tech digital start-ups, but the United Kingdom will close Help to Grow: Digital.
- The United States widened practical training for foreign graduates in cloud and data fields; the United Kingdom opened routes for graduates and scale-ups; routes for founders were plans in Korea and Japan.
- Finance came as vouchers, calls and tax proposals; the United Kingdom will cut its SME R&D relief but extend relief to data and cloud costs and keep larger seed relief.
- Culture received conferences, festivals, rankings and a proclamation rather than programmes, and governments reported mostly outputs and claimed effects, with little evaluation.

Digital entrepreneurship depends on data, computing, rules on data, payments and platforms, and markets reached through platforms that others own; the governance of digital infrastructure is itself a domain of policy (Sussan & Acs, 2017). This report applies the entrepreneurial ecosystem approach to the digital entrepreneurship policy of Korea, Japan, China, the United Kingdom, the United States, France and Germany in 2022. It asks three questions:

1. What did each government set out in each domain of digital entrepreneurship in 2022?
2. How did the policies arise and how were they justified, how far were they valid in theory, and what had they achieved in practice by the end of the year?
3. What were their limitations, and what follows for policy?

The report draws on research on digital ecosystems (Sussan & Acs, 2017), platforms (Cutolo & Kenney, 2021), data regulation (Jia, Jin, & Wagman, 2021), R&D tax relief (Agrawal, Rosell, & Simcoe, 2020), high-growth firms (Mason & Brown, 2013), entrepreneurial culture (Spigel, 2017) and immigrant talent (Hunt, 2011). Sections 2 to 5 review the policies by domain and country, analyse them, set out their limits and draw implications. The report rests on the plans, laws, budgets, reports and announcements of these governments in 2022 (Table 1); quotations from sources not in English are IFEED's translations.

Table 1: Government plans, laws, reports and announcements reviewed

Subject	Issuing bodies	Date and type	Period covered	Description
Korea				
Hyperscale AI pilot	MSIT	16 June 2022 Press release	May–August 2022	Model opened to researchers; SMEs later.
Digital Strategy of Korea	MSIT	28 September 2022 Strategy	September–December 2022	AI Basic Act, visas, exports, unicorns.
Industrial sandbox	MOTIE	20 December 2022 Plan, press release	January 2019–December 2022	Approvals, results and plan.
MSS work plan for 2023	MSS	27 December 2022 Press release	December 2022	Founder funding, COMEUP, academy.
AI Voucher call	MSIT	28 December 2022 Press release	December 2022	Vouchers for buyers of AI.
Japan				
AI strategy	Innovation Strategy Council	22 April 2022 Strategy	April–December 2022	Soft law for AI.
FSA policy	FSA	August 2022 Policy statement	July 2021–December 2022	Fintech desk; conference.
Start-up five-year plan	New Capitalism Council	28 November 2022 Plan	November–December 2022	Visa reform, Web3.0, procurement.
Mobile ecosystem issues	Cabinet Secretariat	22 December 2022 Working group material	April–December 2022	Mobile ecosystem rules studied.
Digital Garden City strategy; tax outline	Cabinet	23 December 2022 Cabinet decisions	December 2022	Talent target; crypto and DX tax.
Draft payment orders	FSA	26 December 2022 Press release	December 2022	Act No. 61; draft orders.
China				
Digital government	State Council	6 June 2022 Opinions	June–December 2022	Digital literacy; training of officials.
Outbound data transfers	CAC	7 July 2022 Rules	September–December 2022	Assessment above volume thresholds.
AI scenario opinions	MOST et al.	29 July 2022 Opinions	July–December 2022	AI start-ups; compute vouchers.
SME digital pilot	MIIT; MOF	15 August 2022 Notice	August–December 2022	Central rewards to service platforms.
SME Digital Service Festival	MIIT	25 August 2022 Notice	August–September 2022	Campaign for SMEs' digital adoption.
Deep synthesis rules	CAC et al.	25 November 2022 Rules	November–December 2022	Identity, labels, filing; from 2023.
Data 20 Articles	CPC Central Committee; State Council	2 December 2022 Opinions	December 2022	Data trading; fair authorisation.

Subject	Issuing bodies	Date and type	Period covered	Description
United Kingdom				
Cryptoasset plan	HM Treasury	4 April 2022 Press release	April 2022	Crypto engagement group.
UK Digital Strategy	DCMS	13 June 2022 Strategy	June–December 2022	Visas; stance on founders.
AI policy paper; AI Action Plan	DCMS; BEIS; Office for AI	18 July 2022 Policy paper, plan	July–December 2022	Non-statutory approach; AI measures.
Autumn Statement	HM Treasury	17 November 2022 Fiscal statement	November 2022	R&D and seed relief; Made Smarter.
Help to Grow: Digital	BEIS	15 December 2022 Press release	December 2022	Closure of the software vouchers.
United States				
STEM degree list	ICE	21 January 2022 Notice	January–December 2022	Cloud and data fields for practical training.
Executive Order 14067	President	9 March 2022 Executive order	March–December 2022	Digital-asset objectives.
CHIPS and Science Act	Congress	9 August 2022 Act	August–December 2022	NSF directorate; fellowships.
Cyber pilot awards	SBA	23 August 2022 Press release	August 2022	Awards to state delegates.
Secure software (M-22-18)	OMB	14 September 2022 Memorandum	September–December 2022	Attestation for software sold to agencies.
Entrepreneurship Month	President	31 October 2022 Proclamation	November 2022	Broadband and online business.
AI Bill of Rights Blueprint	OSTP	October 2022 Framework	October 2022	Five non-binding principles.
France				
French Tech Next40/120	Ministry of the Economy	1 February 2022 Press kit	February 2022	Ranking of leading start-ups.
Electronic invoicing act	Parliament	16 August 2022 Act	August–December 2022	Electronic invoicing; partner platforms.
Cybersecurity projects	MEFIDS	27 October 2022 Press release	February 2021–October 2022	EUR 39 million for 17 projects.
DeepNum20	French Tech Mission	27 October 2022 Press kit	July–October 2022	Deep-tech digital start-ups supported.
Council on digital	MEFIDS	22 November 2022 Press release	November 2022	Consultation on citizens' digital life.
DGE strategy	DGE	2022 Strategy	2022	Digital transition aims; France Num.
Germany				
Google decision	Federal Cartel Office	5 January 2022 Press release	January 2022	Section 19a determination.
Meta decision	Federal Cartel Office	4 May 2022 Press release	May 2022	Section 19a determination.

Subject	Issuing bodies	Date and type	Period covered	Description
Amazon decision	Federal Cartel Office	6 July 2022 Press release	July 2022	Section 19a determination.
Start-up Strategy	Federal Government	28 July 2022 Strategy	July–December 2022	Laboratories, open data, formation.
Skilled immigration key points	Federal Government	30 November 2022 Key points	June–November 2022	Blue Card; IT specialists.
SME innovation report	Federal Government	6 December 2022 Report	January 2021–December 2022	Digital Now, go-digital, competition.

Source: The plans, laws, reports and announcements listed. Periods covered and descriptions: IFEED.

Note: Where a source has several parts, citations name the part (for example, plan or press release) before the page. Abbreviations: AI, artificial intelligence; BEIS, Department for Business, Energy and Industrial Strategy; BMWK, Federal Ministry for Economic Affairs and Climate Action; CAC, Cyberspace Administration of China; CHIPS, Creating Helpful Incentives to Produce Semiconductors; CP, Command Paper; CPC, Communist Party of China; DCMS, Department for Digital, Culture, Media and Sport; DGE, Directorate-General for Enterprises; DX, digital transformation; EU, European Union; FSA, Financial Services Agency; HM, Her or His Majesty's; ICE, U.S. Immigration and Customs Enforcement; ICT, information and communications technology; IPO, initial public offering; IT, information technology; MEFIDS, Ministry of the Economy, Finance and Industrial and Digital Sovereignty; MIIT, Ministry of Industry and Information Technology; MOF, Ministry of Finance; MOST, Ministry of Science and Technology; MOTIE, Ministry of Trade, Industry and Energy; MSIT, Ministry of Science and ICT; MSS, Ministry of SMEs and Startups; NSF, National Science Foundation; OMB, Office of Management and Budget; OSTP, Office of Science and Technology Policy; R&D, research and development; SBA, Small Business Administration; SME, small and medium-sized enterprise; STEM, science, technology, engineering and mathematics; UK, United Kingdom; US and U.S., United States; VAT, value added tax.

2 REVIEW OF DIGITAL ENTREPRENEURSHIP POLICIES

This section reviews the policies of 2022 in six domains of the entrepreneurial ecosystem: regulation and law, finance, culture, support, human capital, and markets.

2.1 REGULATION AND LAW

2.1.1 KOREA: THE INDUSTRIAL SANDBOX AND THE PLANNED AI BASIC ACT

MOTIE's industrial sandbox had approved 327 projects since 2019 and revised the rules behind 51, and MOTIE would consider a special act (MOTIE, 20 December 2022, summary and parts 1–2); the Digital Strategy planned an AI Basic Act (MSIT, 28 September 2022, strategy, p. 68). This was a mixed policy: the sandbox works, but the special act and the basic act were plans.

2.1.2 JAPAN: THE STABLECOIN LAW AND AI STRATEGY 2022

Act No. 61 of 10 June requires dealers in stablecoins to register, from a date within a year (FSA, 26 December 2022, p. 1), and AI Strategy 2022 relied on principles (Integrated Innovation Strategy Promotion Council, 22 April 2022, pp. 2, 40). This was a mixed policy: the law licenses stablecoin dealers and principles spare AI start-ups new compliance costs, but the law's orders were drafts.

2.1.3 CHINA: OUTBOUND DATA MEASURES AND THE DEEP SYNTHESIS PROVISIONS

The CAC's outbound data measures, in force from 1 September, require an assessment for important data and above volume thresholds (CAC, 7 July 2022, Arts. 4, 20); deep synthesis providers will have to verify users, label content and file algorithms from 10 January 2023 (CAC, MIIT, & Ministry of Public Security, 25 November 2022, Arts. 9, 15–19, 25). This was a mixed policy: the thresholds spare small exporters a cost of the kind that, under uniform data rules, has weighed most on young, data-related ventures (Jia et al., 2021), but the deep synthesis duties will bind all providers.

2.1.4 UNITED KINGDOM: THE AI POLICY PAPER

The AI policy paper of 18 July (CP 728) proposed principles on a non-statutory footing, applied by existing regulators (DCMS, BEIS, & Office for AI, 18 July 2022, policy paper, pp. 2, 6, 17–18). This was a mixed policy: the approach spares small firms new compliance costs, but it was still a proposal.

2.1.5 UNITED STATES: EXECUTIVE ORDER 14067 AND THE AI BLUEPRINT

Executive Order 14067 of 9 March set objectives for digital-asset policy (President of the United States, 9 March 2022, secs 2(a)–(b)), and the AI Bill of Rights Blueprint of October was "non-binding" (OSTP, October 2022, pp. 2–3). This was a mixed policy: they spared young firms new duties but set no rules.

2.1.6 FRANCE: THE ELECTRONIC INVOICING LAW

Law No. 2022-1157 of 16 August obliges VAT-liable businesses to invoice electronically through the public portal or partner platforms, phased in from a later date with SMEs last (Parliament of France,

16 August 2022, Art. 26). This was a mixed policy: the law creates a market for invoicing software and will cut small firms' compliance costs, but the duty did not yet apply.

2.1.7 GERMANY: THE REAL-WORLD LABORATORIES ACT AND THE OPEN DATA RIGHT

The Start-up Strategy, adopted on 27 July, announced a Real-World Laboratories Act, AI laboratories with preferential access for start-ups and a right to federal open data (Federal Government of Germany, 28 July 2022, pp. 15–16). This was a mixed policy: the measures would cut start-ups' costs of testing and finding data, but each was announced, not tabled.

2.2 FINANCE: GRANTS, FUNDING AND FINANCIAL INFRASTRUCTURE

2.2.1 KOREA: THE AI VOUCHER PROGRAMME AND FUNDING FOR DIGITAL FOUNDERS

On 28 December MSIT opened its KRW 70 billion AI Voucher call for 2023, for SMEs and other firms adopting AI, after 1,573 applications in 2022 (MSIT, 28 December 2022, paras 1–3, box and attachment 1), and MSS promised up to KRW 100 million for early AI and data founders (MSS, 27 December 2022, part 1(2)). This was a mixed policy: vouchers help SMEs adopt AI and create buyers for AI suppliers, but no project was yet selected and the funding was announced.

2.2.2 JAPAN: THE CRYPTO-ASSET TAX RELIEF AND THE DX INVESTMENT MEASURE

The 23 December tax outline would exempt self-issued, transfer-restricted crypto-assets from mark-to-market tax, and would extend the DX investment tax measure by two years on stricter terms, under "abolition or reduction" (Cabinet of Japan, 23 December 2022, tax outline, pp. 47, 54). This was a mixed policy: the relief would help Web3.0 firms, but it needed legislation, and the DX measure was narrowed.

2.2.3 CHINA: THE FISCAL PILOT FOR SME DIGITAL TRANSFORMATION

Under the notice of 15 August, the central budget rewards platforms that transform pilot SMEs, mainly manufacturers, with up to CNY 6 million each, for about 100 platforms in 2022 (General Offices of MIIT & MOF, 15 August 2022, Parts I and III). This was a mixed policy: central money went into SMEs' digital transformation, but to established manufacturers rather than new ventures.

2.2.4 UNITED KINGDOM: R&D RELIEF FOR DATA AND CLOUD COSTS

The Autumn Statement of 17 November kept the announced rise in seed-investment relief and confirmed R&D relief for data and cloud costs, while from 1 April 2023 the SME deduction will fall from 130 % to 86 % (HM Treasury, 17 November 2022, paras 3.31, 5.52). This was a mixed policy: AI firms and start-ups would gain, but the cut falls on small firms, for which refundable credits have been found to raise R&D (Agrawal et al., 2020).

2.2.5 UNITED STATES: THE CYBERSECURITY FOR SMALL BUSINESS PILOT

On 23 August SBA awarded nearly USD 3 million under its cybersecurity pilot to bodies in Arkansas, Maryland and South Dakota to help nascent and start-up business owners (SBA, 23 August 2022, para. 1 and list of awards). This was a mixed policy: the pilot pays for start-ups' cybersecurity help, but it was small and paid bodies chosen by the states, not firms.

2.2.6 FRANCE: THE CYBERSECURITY CALLS OF THE INVESTMENT PLAN

On 27 October the government announced EUR 39 million for 17 cybersecurity projects from five calls (MEFIDS, 27 October 2022, pp. 1–2). This was a mixed policy: the calls funded security projects by competition, but part of the money went to training and shared data rather than to firms.

2.2.7 GERMANY: THE PLANNED FUTURE FINANCING ACT

The Start-up Strategy named a planned Future Financing Act as a key step to ease start-ups' initial public offerings (Federal Government of Germany, 28 July 2022, p. 4). This was a mixed policy: the act would open capital markets to growth firms, but it was a plan.

2.3 CULTURE

2.3.1 KOREA: THE COMEUP START-UP FESTIVAL

MSS planned to grow COMEUP, its start-up festival, into a top-five global start-up event (MSS, 27 December 2022, part 1(1)). This was a mixed policy: a festival builds networks through which, it has been argued, an entrepreneurial culture is reproduced (Spigel, 2017), but its growth was a plan.

2.3.2 JAPAN: FSA'S FINTECH CONFERENCE AND THE START-UP DECLARATION

FSA held a fintech conference with firms and officials from 29 to 31 March (FSA, August 2022, p. 176), and the Start-up Development Five-year Plan declared 2022 the "first year of start-up creation" (New Capitalism Realisation Council, 28 November 2022, p. 1). This was a mixed policy: the conference builds relations through which, it has been argued, a culture is sustained (Spigel, 2017), but the declaration set no programme.

2.3.3 CHINA: THE NATIONAL SME DIGITAL SERVICE FESTIVAL

MIIT's General Office scheduled an SME Digital Service Festival from 28 August to 28 September for SMEs "unwilling, not knowing how, afraid or unable" to go digital (General Office of MIIT, 25 August 2022, Parts I and III). This was a mixed policy: it brought providers to reluctant SMEs, but promoted adoption, not founding, and reported no results.

2.3.4 UNITED KINGDOM: THE UK DIGITAL STRATEGY AND CRYPTO ENGAGEMENT

The UK Digital Strategy of 13 June called founders' and investors' success "a national success, to be applauded, encouraged and emulated – not criticised" (DCMS, 13 June 2022, foreword), and on 4 April HM Treasury announced a Cryptoasset Engagement Group (HM Treasury, 4 April 2022, section on other measures). This was a mixed policy: the stance builds role models (Spigel, 2017) but set no programme, and the group's work was not reported.

2.3.5 UNITED STATES: NATIONAL ENTREPRENEURSHIP MONTH

Proclamation 10487 of 31 October declared November National Entrepreneurship Month, noting that broadband would let Americans anywhere run a business online (President of the United States, 31 October 2022, p. 66525). This was a poor policy: the proclamation set no programme and

addressed entrepreneurs in general, touching digital firms only through broadband, so it built no networks of digital founders (Spigel, 2017).

2.3.6 FRANCE: FRENCH TECH NEXT40/120 AND THE COUNCIL ON DIGITAL

On 1 February the government announced the 2022 French Tech Next40/120 ranking of leading start-ups (Ministry of the Economy, Finance and Recovery, 1 February 2022, p. 1), and on 22 November launched the digital strand of the National Council for Renewal, on citizens' digital life (MEFIDS, 22 November 2022, p. 1). This was a mixed policy: the ranking makes founders role models (Spigel, 2017), but the council addressed citizens.

2.3.7 GERMANY: THE DIGITAL INNOVATIONS FOUNDING COMPETITION

The BMWK's founding competition "Digital Innovations" honoured 21 teams in each round of 2021 (Federal Government of Germany, 6 December 2022, p. 3), and the Start-up Strategy called for appreciation of founders (Federal Government of Germany, 28 July 2022, p. 2). This was a mixed policy: the competition honours digital founders (Spigel, 2017), but its results were from 2021 and the appreciation was a statement.

2.4 SUPPORT: INSTITUTIONAL SUPPORT

2.4.1 KOREA: THE HYPERSCALE AI PILOT AND THE DIGITAL VENTURE VALLEY

In June MSIT opened a hyperscale AI model to research bodies, considering SMEs later (MSIT, 16 June 2022, paras 1, 5), and MSS planned a virtual "digital venture valley" (MSS, 27 December 2022, part 1(2)). This was a mixed policy: AI models are inputs that, it has been argued, a digital ecosystem must open (Sussan & Acs, 2017), but the pilot served researchers and the valley was a plan.

2.4.2 JAPAN: THE FINTECH SUPPORT DESK AND THE WEB3.0 DESK

FSA's FinTech Support Desk received 306 consultations in the 2021 administrative year (FSA, August 2022, p. 174), and a Web3.0 desk for local governments was planned (Cabinet of Japan, 23 December 2022, strategy, p. 42). This was a mixed policy: the fintech desk eases dealings with financial rules, but the Web3.0 desk was a plan and served local governments.

2.4.3 CHINA: THE AI SCENARIO OPINIONS

MOST's opinions of 29 July named AI start-ups for support and encouraged localities to cut AI firms' infrastructure costs, including through innovation vouchers (MOST et al., 29 July 2022, items 8, 15). This was a mixed policy: the opinions named AI start-ups, but set no budget and left the vouchers to localities.

2.4.4 UNITED KINGDOM: HELP TO GROW: DIGITAL AND MADE SMARTER ADOPTION

BEIS announced on 15 December that Help to Grow: Digital, designed for 100,000 SMEs, will close to new applications on 2 February 2023 after fewer than 1,000 vouchers were redeemed (BEIS, 15 December 2022, paras 1–4), and the Autumn Statement announced the extension of Made Smarter Adoption for smaller manufacturers to the East Midlands (HM Treasury, 17 November 2022, para.

3.29). This was a mixed policy: Made Smarter Adoption ran, but the main SME software scheme will close.

2.4.5 UNITED STATES: THE NSF DIRECTORATE AND INNOVATION ENGINES

The CHIPS and Science Act of 9 August established NSF's Directorate for Technology, Innovation, and Partnerships and provided for Regional Innovation Engines (United States Congress, 9 August 2022, secs 10381, 10388). This was a mixed policy: the Act gives research translation a statutory base, but the directorate and the engines serve research translation and regions rather than start-ups.

2.4.6 FRANCE: THE DEEPNUM20 PROGRAMME AND FRANCE NUM

On 27 October the French Tech Mission named the first DeepNum20 cohort of deep-tech digital start-ups, each given a start-up manager and access to State services (French Tech Mission, 27 October 2022, pp. 6–9); France Num had delivered nearly 150,000 actions for small firms (DGE, 2022, p. 9). This was a good policy: DeepNum20 and France Num reached their firms directly.

2.4.7 GERMANY: DIGITAL NOW, GO-DIGITAL AND THE DIGITAL HUB INITIATIVE

Digital Now had approved more than 2,000 grants by the end of 2021; go-digital, which nearly 94 % of SMEs surveyed in its evaluation said had changed how they digitalised, was renewed from January 2022; and the Digital Hub Initiative linked start-ups with the Mittelstand (Federal Government of Germany, 6 December 2022, pp. 4–5). Formation within 24 hours was an aim (Federal Government of Germany, 28 July 2022, p. 8). This was a mixed policy: adoption grants and hubs ran, but fast formation was an aim.

2.5 HUMAN CAPITAL: EDUCATION AND TRAINING, AND LABOUR

2.5.1 KOREA: THE VISA PLANS OF THE DIGITAL STRATEGY

The Digital Strategy planned an intern visa for foreign students, a negative-list visa for top talent and longer stays for technology founders (MSIT, 28 September 2022, strategy, p. 38), and MSS a larger venture academy (MSS, 27 December 2022, part 1(2)). This was a mixed policy: the plans would widen the talent pool, but immigrants, who, it has been found, start companies at higher rates than similar natives (Hunt, 2011), gained no new route in the year.

2.5.2 JAPAN: THE DIGITAL TALENT TARGET AND THE START-UP VISA REFORM

The Digital Garden City Nation strategy aims to train 2.3 million digital personnel in five years (Cabinet of Japan, 23 December 2022, strategy, p. 31), and the Start-up Development Five-year Plan promised a longer stay under the start-up visa, with private bodies allowed to confirm founders' plans (New Capitalism Realisation Council, 28 November 2022, p. 9). This was a mixed policy: the reform would widen a founder route, but the reform and the target were plans.

2.5.3 CHINA: THE DIGITAL LITERACY SYSTEM OF THE DIGITAL GOVERNMENT OPINIONS

The State Council's opinions on digital government of 6 June called for a national digital literacy system and training of officials, naming no founders (State Council, 6 June 2022, Part VIII, item (3)).

This was a poor policy: literacy aims reach neither founders nor immigrants, who, evidence from the United States shows, start companies at higher rates than similar natives (Hunt, 2011).

2.5.4 UNITED KINGDOM: HIGH POTENTIAL INDIVIDUAL AND SCALE-UP ROUTES

The Digital Strategy listed a High Potential Individual route for top graduates from 30 May and a Scale-up route from 22 August, and announced a revised Innovator route (DCMS, 13 June 2022, section 3.6). This was a mixed policy: the routes widen digital firms' hiring pool, but the founder route's revision had no confirmed start.

2.5.5 UNITED STATES: THE STEM DEGREE LIST AND THE NSF FELLOWSHIPS

On 21 January the Department of Homeland Security added 22 fields, including cloud computing and data science, to those giving F-1 graduates 24 more months of practical training (ICE, 21 January 2022, pp. 3317, 3319), and NSF entrepreneurial fellowships had funds authorised from fiscal year 2023 (United States Congress, 9 August 2022, sec. 10392(a), (e)). This was a mixed policy: data graduates stay available to digital firms, but the fellowships awaited funds.

2.5.6 FRANCE: THE CYBERSECURITY SKILLS PROJECTS

The cybersecurity projects of 27 October include a skills diagnostic for 37,000 expected jobs (MEFIDS, 27 October 2022, pp. 1, 3), and DeepNum20 start-ups gained access to the French Tech Visa (French Tech Mission, 27 October 2022, p. 9). This was a mixed policy: the projects address a shortage of security skills, but no new route opened for foreign talent or founders.

2.5.7 GERMANY: THE KEY POINTS ON SKILLED-WORKER IMMIGRATION

On 30 November the Cabinet adopted key points to lower Blue Card salary thresholds and ease rules for IT specialists, with bills due in early 2023; online Blue Card applications had begun in June (Federal Government of Germany, 30 November 2022, pp. 2, 5–6, 18). This was a mixed policy: hiring would ease, but the reform was a set of key points without a founder route.

2.6 MARKET: MARKET AND INDUSTRY SUPPORT, IPO AND EXIT

2.6.1 KOREA: THE DIGITAL EXPORT PLANS

The Digital Strategy planned, from 2023, a digital export pioneer group and digital offices in Silicon Valley, Brussels and Shanghai (MSIT, 28 September 2022, strategy, p. 39). This was a mixed policy: the plans would open foreign markets to digital ventures, but they were plans.

2.6.2 JAPAN: THE DIGITAL MARKETPLACE AND THE MOBILE ECOSYSTEM REVIEW

The Start-up Development Five-year Plan provided for a digital marketplace of IT services, start-ups' among them, to be piloted in fiscal year 2023 (New Capitalism Realisation Council, 28 November 2022, p. 15), and ex-ante rules for mobile ecosystems were under study (Cabinet Secretariat of Japan, 22 December 2022, pp. 2, 25). This was a mixed policy: a marketplace would give start-ups public buyers, but the marketplace and the rules were not yet in place.

2.6.3 CHINA: DATA TRADING VENUES UNDER THE DATA 20 ARTICLES

The Data 20 Articles of 2 December asked platforms to authorise data fairly with small firms and called for rules on data trading (CPC Central Committee & State Council, 2 December 2022, Part II, art. (5), and Part III, arts. (9)–(10)). This was a mixed policy: fair access to platform data would help small firms, but the rules were still to come.

2.6.4 UNITED KINGDOM: THE DIGITAL MARKETS BILL AND THE AI MARKETPLACE

The Autumn Statement said that the Digital Markets, Competition and Consumer Bill would come in the third parliamentary session (HM Treasury, 17 November 2022, paras 3.32, 5.71), and the AI Marketplace for public buyers was renewed (DCMS et al., 18 July 2022, action plan, Pillar Two). This was a mixed policy: the bill addresses the power that, it has been argued, platform owners hold over dependent firms (Cutolo & Kenney, 2021), but it had not been introduced.

2.6.5 UNITED STATES: MEMORANDUM M-22-18 ON SECURE SOFTWARE

OMB memorandum M-22-18 of 14 September requires agencies to obtain secure-development attestations from software producers within 270 days for critical software and 365 days for all software (OMB, 14 September 2022, pp. 3–6). This was a mixed policy: attestation makes software sold to agencies safer, but it is a fixed cost that weighs most on small software firms.

2.6.6 FRANCE: PARTNER INVOICING PLATFORMS AND THE EU DIGITAL MARKETS RULES

The invoicing law lets registered partner platforms carry invoices beside the public portal (Parliament of France, 16 August 2022, Art. 26), and the DGE aimed to carry the EU's digital markets rules further in French law (DGE, 2022, p. 10). This was a mixed policy: partner status creates a market for invoicing software firms, but the duty behind it applies only from a later date and the EU follow-up was an aim.

2.6.7 GERMANY: THE SECTION 19A DECISIONS AND THE PROCUREMENT PLANS

The Federal Cartel Office placed Alphabet (announced on 5 January), Meta (4 May) and Amazon (decided on 5 July) under section 19a abuse control for five years, for Meta from the date the decision becomes final (Federal Cartel Office, 5 January 2022, paras 1, 9; Federal Cartel Office, 4 May 2022, paras 1, 9; Federal Cartel Office, 6 July 2022, paras 1, 7); the Start-up Strategy planned an e-marketplace for innovative procurement and AI vouchers (Federal Government of Germany, 28 July 2022, pp. 13, 15). The section 19a decisions were a good policy: they put under special abuse control the platform owners on which, it has been argued, digital firms depend (Cutolo & Kenney, 2021). The procurement plans were a mixed policy: they would give start-ups buyers but were plans.

3 ANALYSIS OF THE POLICIES

3.1 CONTEXTUAL ANALYSIS: ORIGINS, BACKGROUND AND RATIONALE OF THE POLICIES

Korea's new government made the Digital Strategy a national task (MSIT, 28 September 2022, strategy, p. i), and the Autumn Statement of 17 November reversed nearly all the United Kingdom's Growth Plan (HM Treasury, 17 November 2022, p. 6). Each government framed the problem in its own way (Table 2).

Table 2: How the governments represented the problem

Country	Problem represented	Presupposition	Left unsaid
Korea	Too few digital unicorns (MSIT, 28 September 2022, strategy, p. 38)	Visas and vouchers scale firms	Buyers for start-ups
Japan	AI rules must not unduly constrain the economy (Integrated Innovation Strategy Promotion Council, 22 April 2022, p. 40)	Principles suffice	Start-ups' computing
China	SMEs are unwilling, not knowing how, afraid or unable to go digital (General Office of MIIT, 25 August 2022, Part I)	State pilots spread technology	Founders
United Kingdom	Unclear AI law hurts smaller businesses (DCMS et al., 18 July 2022, policy paper, key challenges)	Regulators can govern AI	A voucher successor; compute
United States	"Same business, same risks, same rules" (President of the United States, 9 March 2022, sec. 2(b))	Executive direction suffices	Digital founders
France	France must reach world-champion rank in cybersecurity (MEFIDS, 27 October 2022, p. 1)	Champions secure it	Other founders
Germany	Start-ups need easier initial public offerings (Federal Government of Germany, 28 July 2022, p. 4)	New laws free start-ups	When laws will come

Source: IFEED analysis of the sources cited.

3.2 THEORETICAL ANALYSIS: THEORETICAL VALIDITY AND RELEVANCE

Digital ventures, it has been argued, arise where data and rules let them reach markets (Sussan & Acs, 2017), and uniform EU data rules cut investment most in young, data-related ventures (Jia et al., 2021) (Table 3).

Table 3: Theoretical validity of the main policies

Theory	Claim	Policies consistent with it	Policies in tension with it
Digital ecosystems (Sussan & Acs, 2017)	Open data let ventures reach markets	Open data right (Germany); DeepNum20 (France)	Research-only AI pilot (Korea); unbudgeted vouchers (China)
Platforms (Cutolo & Kenney, 2021)	Owners hold power over dependent firms	Section 19a decisions (Germany)	Rules pending (United Kingdom, Japan)
Data regulation (Jia et al., 2021)	Uniform rules hold back young ventures	Volume thresholds (China)	Deep synthesis duties for all providers (China)
Immigrant talent (Hunt, 2011)	Immigrants start more firms than natives	STEM list (United States); new routes (United Kingdom)	Founder visas planned (Korea, Japan)

Source: IFEED assessment of the policies reviewed in Section 2, against the sources named in the first column.

3.3 PRACTICAL ANALYSIS: PERFORMANCE, CONTRIBUTION AND IMPACT

Much of the year's policy had not reached firms by its end (Table 4). Governments reported mostly outputs; effects were claimed or surveyed, as MOTIE's KRW 163.1 billion of sales and 866 jobs at sandbox firms (MOTIE, 20 December 2022, summary) and the go-digital evaluation (Federal Government of Germany, 6 December 2022, p. 5). Table 5 rates a policy good where all its measures were in force or launched and reached their firms; mixed where they sat beside pending ones or reached others, or only pending measures for digital firms were set out; and poor where nothing in force reached digital firms or founders and the rest aimed elsewhere.

Table 4: Status of the main measures at the end of 2022

Country	In force or launched in 2022	Adopted, in force from 2023	Pending or announced
Korea	Sandbox; AI pilot; AI Voucher call	—	AI Basic Act; sandbox act; founder funding; COMEUP; venture valley; visas; exports
Japan	AI Strategy 2022; FinTech Support Desk; fintech conference	Stablecoin law (Act No. 61)	Draft orders; tax outline; Web3.0 desk; visa reform; talent target; marketplace; platform rules
China	Outbound data measures; SME pilot; festival	Deep synthesis provisions	Data 20 rules; computing vouchers; literacy system
United Kingdom	High Potential Individual and Scale-up routes; Made Smarter Adoption; AI Marketplace	Closure of Help to Grow: Digital	AI approach; Digital Markets Bill; R&D relief; Made Smarter extension; Innovator route; crypto engagement group
United States	Executive Order 14067; AI Blueprint; M-22-18; STEM list; SBA pilot; CHIPS and Science Act; Proclamation 10487	NSF fellowships (fiscal year 2023)	—
France	Cybersecurity calls; DeepNum20; Next40/120; France Num; council on digital	Invoicing duty (phased in later)	Follow-up of EU digital rules
Germany	Section 19a decisions; Digital Now; go-digital; hubs; competition; online Blue Card	—	Laboratories Act; open data; Future Financing Act; immigration reform; 24-hour formation; procurement plans

Source: IFEED compilation from the sources cited in Section 2.

Table 5: Assessment of the policies by domain and country

Domain	Korea	Japan	China	United Kingdom	United States	France	Germany
Regulation and Law	Mixed	Mixed	Mixed	Mixed	Mixed	Mixed	Mixed
Finance	Mixed	Mixed	Mixed	Mixed	Mixed	Mixed	Mixed
Culture	Mixed	Mixed	Mixed	Mixed	Poor	Mixed	Mixed
Support	Mixed	Mixed	Mixed	Mixed	Mixed	Good	Mixed
Human Capital	Mixed	Mixed	Poor	Mixed	Mixed	Mixed	Mixed
Market	Mixed	Mixed	Mixed	Mixed	Mixed	Mixed	Mixed

Source: IFEED assessment of the policies reviewed in Section 2. Where a subsection assessed its policies differently, Mixed is shown.

4 POLICY LIMITATIONS

4.1 POLICY DEBATES, CONTESTED POINTS AND ISSUES

Of the six points of contention in Table 6, the sharpest were whether platform power should be checked case by case, as in Germany, or by rules set in advance, as the United Kingdom planned and Japan studied, and whether adoption subsidies give value for money, which the United Kingdom questioned in closing its voucher scheme while extending Made Smarter Adoption.

Table 6: Debates and contested points in the policies of 2022

Issue	What the governments set out	Point at issue
Data duties	Volume thresholds (China); open data right announced (Germany)	Protection of data against young firms' fixed costs
AI rules	Principles (Japan, United Kingdom); blueprint (United States); deep synthesis rules (China); basic act planned (Korea)	Legal certainty against compliance costs
Crypto-assets	Stablecoin law (Japan); executive order (United States); crypto-asset tax relief proposed (Japan)	Licensed innovation against investor protection
Platform power	Section 19a decisions (Germany); bill planned (United Kingdom); rules studied (Japan)	Case-by-case enforcement against rules in advance
SME adoption	Fiscal pilot (China); grants (Germany); France Num (France); AI vouchers (Korea); vouchers closed, Made Smarter extended (United Kingdom)	Subsidies against value for money
Talent	Practical training (United States); graduate routes (United Kingdom); founder visas planned (Korea, Japan); key points (Germany)	Migration control against the supply of talent

Source: IFEED analysis of the sources cited in Section 2.

4.2 THEORETICAL LIMITATIONS OF THE POLICIES

First, the policies treated digital technology as something to adopt, not as the infrastructure on which, it has been argued, new ventures arise (Sussan & Acs, 2017); support and culture aimed mostly at users of digital tools, and few reached founders. Second, they selected: Korea aimed to raise its digital unicorns from 23 to 100 within five years (MSIT, 28 September 2022, strategy, p. 38), China supported "little giants" and unicorns in AI scenarios (MOST et al., 29 July 2022, item 8), and France's DeepNum20 picked start-ups by jury, although high-growth firms, it has been argued, vary too much for support to target particular technologies (Mason & Brown, 2013).

Third, duties were uniform. China's deep synthesis provisions will apply to small firms as to large ones, although uniform data rules have been found to fall hardest on new, data-related ventures (Jia et al., 2021); the US attestation for software sold to agencies weighs most on small sellers; and the United Kingdom's cut in SME R&D relief will fall on small firms, for which refundable credits have been found to raise R&D (Agrawal et al., 2020). Finally, evaluation was weak: the effects reported were claimed by ministries or surveyed users, and only the United Kingdom explained a closure by its results.

5 POLICY IMPLICATIONS

First, data and computing should reach start-ups by rule. Korea should open its hyperscale AI pilot to SMEs and start-ups as it planned to consider, Germany should table its right to open data, China should fund the computing vouchers that it encouraged localities to offer, and Japan and the United Kingdom should state how start-ups will reach public computing.

Second, digital duties should be scaled to the size of firms. China's thresholds for data exports show how; China's deep synthesis duties and the US attestation for software sold to agencies should spare or phase in small firms, and Korea and Germany should enact their sandbox laws so that young firms can test products under exemption.

Third, platform rules should be set in law. The United Kingdom should introduce its digital markets bill and Japan should decide on rules for mobile ecosystems; Germany's section 19a decisions show that a competition authority can act against platform owners once the law allows it.

Fourth, governments should act as paying first customers. Japan should pilot its digital marketplace, Germany should open its e-marketplace for innovative procurement, and governments should publish every year the share of their digital purchases made from young firms.

Finally, talent rules and evaluation should serve digital founders. Korea and Japan should enact their visa changes for founders, the United Kingdom should confirm the start of its revised Innovator route, and Germany should add a route for founders to its immigration reform. Governments should publish the take-up and effects of their digital programmes, as the United Kingdom did for take-up in closing its voucher scheme and Germany began to do for effects in reporting the evaluation of go-digital, and the United Kingdom should keep relief for small firms' R&D while extending it to data and cloud costs.

REFERENCES

Agrawal, A., Rosell, C., & Simcoe, T. (2020). Tax credits and small firm R&D spending. *American Economic Journal: Economic Policy, 12*(2), 1–21. http://dx.doi.org/10.1257/pol.20140467

Cabinet of Japan. (2022, December 23). *Digital Garden City Nation comprehensive strategy*, with the *Outline of the tax reform for fiscal year 2023* of the same day [Cabinet decisions, in Japanese].

Cabinet Secretariat of Japan. (2022, December 22). *Digital Market Competition Council working group: Issues to be examined by topic* [Working group material, in Japanese].

Central Committee of the Communist Party of China & State Council of the People's Republic of China. (2022, December 2). *Opinions on building a basic data system to better bring the role of data factors into play* [Opinions, in Chinese].

Cutolo, D., & Kenney, M. (2021). Platform-dependent entrepreneurs: Power asymmetries, risks, and strategies in the platform economy. *Academy of Management Perspectives, 35*(4), 584–605. http://dx.doi.org/10.5465/amp.2019.0103

Cyberspace Administration of China. (2022, July 7). *Measures for the security assessment of outbound data transfers* (Order No. 11) [Departmental rules, in Chinese].

Cyberspace Administration of China, Ministry of Industry and Information Technology, & Ministry of Public Security. (2022, November 25). *Provisions on the administration of deep synthesis in internet information services* (Order No. 12) [Departmental rules, in Chinese].

Department for Business, Energy and Industrial Strategy. (2022, December 15). *Final opportunity for businesses to access Help to Grow: Digital scheme* [Press release].

Department for Digital, Culture, Media and Sport. (2022, June 13). *UK Digital Strategy* [Policy paper].

Department for Digital, Culture, Media and Sport, Department for Business, Energy and Industrial Strategy, & Office for Artificial Intelligence. (2022, July 18). *Establishing a pro-innovation approach to regulating AI* (CP 728), with the *National AI Strategy: AI Action Plan* of the same day [Policy paper and action plan].

Directorate-General for Enterprises. (2022). *Strategy of the Directorate-General for Enterprises* [Strategy, in French].

Federal Cartel Office. (2022, January 5). *Alphabet/Google is a test case for the new supervision of large digital companies* [Press release, in German].

Federal Cartel Office. (2022, May 4). *New rules apply to Meta (formerly Facebook): Federal Cartel Office determines paramount significance for competition across markets* [Press release, in German].

Federal Cartel Office. (2022, July 6). *Amazon now subject to stricter rules: Federal Cartel Office determines its paramount significance for competition across markets* [Press release, in German].

Federal Government of Germany. (2022, July 28). *Start-up strategy of the Federal Government* (Printed paper 20/3063) [Strategy, in German].

Federal Government of Germany. (2022, November 30). *Key points on skilled-worker immigration from third countries* [Cabinet key points, in German].

Federal Government of Germany. (2022, December 6). *Report on the programmes for innovation and technology support in the Mittelstand* (Printed paper 20/4979) [Report, in German].

Financial Services Agency. (2022, August). *Financial administration policy for the 2022 administrative year* [Policy statement, in Japanese].

Financial Services Agency. (2022, December 26). *Publication of the draft orders for the 2022 amendment of the Payment Services Act* [Press release, in Japanese].

French Tech Mission. (2022, October 27). *French Tech DeepNum20 programme: Announcement of the laureates of the first edition* [Press kit, in French].

General Office of the Ministry of Industry and Information Technology. (2022, August 25). *Notice on holding the SME Digital Service Festival* [Notice, in Chinese].

General Offices of the Ministry of Industry and Information Technology & Ministry of Finance. (2022, August 15). *Notice on launching the pilot of fiscal support for the digital transformation of SMEs* [Notice, in Chinese].

HM Treasury. (2022, April 4). *Government sets out plan to make UK a global cryptoasset technology hub* [Press release].

HM Treasury. (2022, November 17). *Autumn Statement 2022* (CP 751) [Fiscal statement].

Hunt, J. (2011). Which immigrants are most innovative and entrepreneurial? Distinctions by entry visa. *Journal of Labor Economics, 29*(3), 417–457. http://dx.doi.org/10.1086/659409

Integrated Innovation Strategy Promotion Council. (2022, April 22). *AI strategy 2022* [Council decision, in Japanese].

Jia, J., Jin, G. Z., & Wagman, L. (2021). The short-run effects of the General Data Protection Regulation on technology venture investment. *Marketing Science, 40*(4), 661–684. http://dx.doi.org/10.1287/mksc.2020.1271

Mason, C., & Brown, R. (2013). Creating good public policy to support high-growth firms. *Small Business Economics, 40*(2), 211–225. http://dx.doi.org/10.1007/s11187-011-9369-9

Ministry of Science and ICT. (2022, June 16). *Pilot project using hyperscale AI starts, to build a hyperscale AI ecosystem* [Press release, in Korean].

Ministry of Science and ICT. (2022, September 28). *Digital Strategy of Korea* [Press release and strategy, in Korean].

Ministry of Science and ICT. (2022, December 28). *Concentrated support for AI chips, health care and other strategic fields to back the Digital Strategy of Korea: Call for the 2023 AI Voucher programme* [Press release, in Korean].

Ministry of Science and Technology, Ministry of Education, Ministry of Industry and Information Technology, Ministry of Transport, Ministry of Agriculture and Rural Affairs, & National Health Commission. (2022, July 29). *Guiding opinions on accelerating scenario innovation to promote high-quality development through high-level application of AI* [Opinions, in Chinese].

Ministry of SMEs and Startups. (2022, December 27). *A global start-up powerhouse for the digital economy era: Work plan for 2023* [Press release, in Korean].

Ministry of the Economy, Finance and Industrial and Digital Sovereignty. (2022, October 27). *New laureates of the investment plan: The government unveils 17 major projects to raise France to world-champion rank in cybersecurity* [Press release, in French].

Ministry of the Economy, Finance and Industrial and Digital Sovereignty. (2022, November 22). *Launch of the National Council for Renewal "Digital"* [Press release, in French].

Ministry of the Economy, Finance and Recovery. (2022, February 1). *French Tech Next40/120: The 2022 cohort* [Press kit, in French].

Ministry of Trade, Industry and Energy. (2022, December 20). *Upgrading the regulatory sandbox to new-industry trends: Regulatory sandbox development plan* [Press release, in Korean].

New Capitalism Realisation Council. (2022, November 28). *Start-up development five-year plan* [Plan, in Japanese].

Office of Management and Budget. (2022, September 14). *Enhancing the security of the software supply chain through secure software development practices* (Memorandum M-22-18) [Memorandum].

Office of Science and Technology Policy. (2022, October). *Blueprint for an AI bill of rights: Making automated systems work for the American people* [Framework].

Parliament of France. (2022, August 16). *Law No. 2022-1157 amending the finance law for 2022* (Art. 26) [Act, Official Journal, in French].

President of the United States. (2022, March 9). *Ensuring responsible development of digital assets* (Executive Order 14067). *Federal Register, 87*, 14143–14152.

President of the United States. (2022, October 31). *National Entrepreneurship Month, 2022* (Proclamation 10487). *Federal Register, 87*, 66525–66526.

Small Business Administration. (2022, August 23). *SBA Administrator Guzman announces grant awardees for new pilot program to bolster cybersecurity infrastructure for emerging small businesses* [Press release].

Spigel, B. (2017). The relational organization of entrepreneurial ecosystems. *Entrepreneurship Theory and Practice, 41*(1), 49–72. http://dx.doi.org/10.1111/etap.12167

State Council of the People's Republic of China. (2022, June 6). *Guiding opinions on strengthening the building of digital government* [Opinions, in Chinese].

Sussan, F., & Acs, Z. J. (2017). The digital entrepreneurial ecosystem. *Small Business Economics, 49*(1), 55–73. http://dx.doi.org/10.1007/s11187-017-9867-5

United States Congress. (2022, August 9). *CHIPS and Science Act* (Public Law 117-167) [Act].

U.S. Immigration and Customs Enforcement. (2022, January 21). *Update to the Department of Homeland Security STEM Designated Degree Program List* [Notice]. *Federal Register, 87*, 3317–3321.